

Siddharth Yadav

571-461-9012 | siddharthyadav555@gmail.com | [linkedin.com/in/siddharthydv](https://www.linkedin.com/in/siddharthydv) | github.com/Siddhart-Yadav

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Bash, HTML, CSS, YAML

Frameworks & Tools: React, Next.js, Node.js, Express.js, FastAPI, REST API, GraphQL, PostgreSQL, MongoDB, Supabase, Firebase, Redis, Kafka, Mongoose, MySQL, OAuth, JWT, SSO, CI/CD, Git, Agile, OpenAPI

Cloud & DevOps: AWS (EC2, Lambda, S3, CloudWatch, RDS, API Gateway, SQS, EKS), Vercel, GCP, Docker, Kubernetes, Linux

AI/ML & Automation: Claude Code, Cursor, LangGraph, LangChain, PyTorch, LLM's, Groq, OpenAI API, Anthropic API, Gemini, Semgrep, MCP

EXPERIENCE

Commonwealth Cyber Initiative

Jan 2025 – Present

Researcher

Arlington, VA

- Implemented an LLM agent system on LangGraph to triage security alerts and auto-generate detection rules, reducing mean time to detect by **50%**;
- Built reusable RAG ingestion and retrieval pipelines on LlamaIndex and ChromaDB supporting vector and knowledge-graph modes, citation-aware querying, and incremental hash-based re-indexing, exposed as composable tools any agent in the platform can call.
- Designed a tool-first agent platform on LangGraph to enable composition of specialist AI assistants from a shared tool registry (RAG, SIEM, formal verification, web search, fine-tuning), with a provider-abstracted LLM layer and MCP server exposure for AI-IDE integration.

American Chemical Society

May 2025 – Aug 2025

Software Engineering Intern

Washington, DC

- Architected an agentic workflow on ACS Publications integration platform using **LangGraph's ReAct loop** orchestrating multi-step API integrations across AWS services and third-party SaaS vendors, reducing edge-case integration failure resolution time by **70%**.
- Engineered an agentic **GitHub PR analytics** system to eliminate manual identification of high-risk PRs by automating data collection, which led to **85%** reduction in review effort.
- Set up a **GitHub Actions CI/CD pipeline** for ACS Pubs repos, increasing deployment frequency from weekly to over **10 deployments per day**.

TATA AIG General Insurance

Sep 2022 – July 2024

Software Engineer

Gurugram, India

- Developed the backend services of a highly scalable portal for creating and modifying policies using **Lambda, RDS, NodeJS, S3**, achieving a **90%** improvement in TAT.
- Developed a cross-organization report and access management platform using **ReactJS, EC2 and ExpressJS**, reducing the time required for generating and sharing reports by **80%**.
- Successfully integrated the **legacy Genesis Configurator** API into the backend system, modernizing the infrastructure and improving system performance.
- Implemented different product API's by adding authorization and rate limiting using **AWS API Gateway, VPC, Redis**, directly enabling new vendor integrations and partnerships.

Infosys Limited

Nov 2020 – Aug 2022

System Engineer

Mysore, India

- Built a data sync pipeline for a banking client to sync customer and loan records across **RDS, S3, and downstream systems**, implementing incremental batch processing with offset-based pagination and automated status tracking.
- Designed and implemented a retrying microservice for syncing cases to downstream leveraging **Step Functions, and Redis**, reducing failure rate by **70%** and ensuring data consistency.
- Mentored new employees** on clean code and best practices, conducted code reviews, and led daily stand ups.

PROJECTS

AI Code Review Agent | Python, LangGraph, FastAPI, Next.js, Docker, Groq/Llama 3.3

- Built a **multi-agent code review system** using LangGraph with 4 parallel specialist agents (security, performance, style, test coverage) and a **multi-provider LLM abstraction** supporting Groq, Gemini, OpenAI, and Anthropic.
- Implemented smart diff chunking with priority-based file ranking, achieving **100% context precision, 86% F1 score**, and **61% token savings**, validated against real freeCodeCamp PRs.

Bug Localization and Repair Using LLM's | PyTorch, Llama 3B, GPT-4

- Built a **dual-model** system using fine-tuned **Llama 3B Instruct** for automated bug localization and repair, benchmarked against GPT-4 and Qwen Coder 32B.
- Augmented the QuixBugs dataset** with syntactic and logical single-line bugs, and demonstrated that targeted fine-tuning can outperform larger general-purpose LLMs, while improving accuracy.

EDUCATION

Virginia Tech

M.S. in Computer Science

Washington DC Baltimore Area, US

Aug 2024 – May 2026

Galgotias University

B.Tech in Electronics and Communication Engineering

Greater Noida, India

Aug 2016 – Jul 2020